

NN from Scratch
Embedded AI – Assignment 2
EMBD X424
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https://www.github.com/electrickitty2/embedx424_assignment_2.git

Assignment

- Implement the JPYNB checkerboard example
- Fill out the missing parts and visualize
- Show your results, screenshot, video, presentation
- <https://drive.google.com/file/d/1y3ZtLQ9fv7iDVJlfey4jkDo9avNbImUQ/view?usp=sharing>

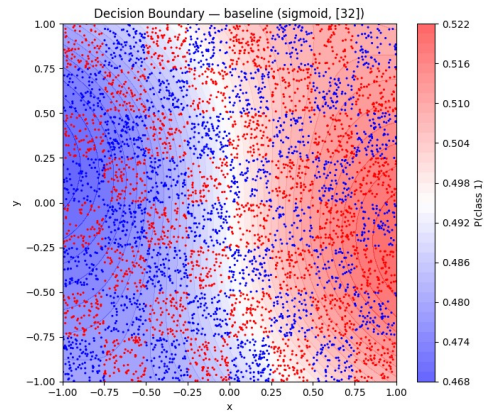
Extra Credit – Circle Classification

- Instead of a checkerboard, classify points based on whether they lie inside a circle:
- Input: (x, y)
- Output: 1 if inside circle, else 0
- Decision boundary: $x^2 + y^2 < r^2$
- Questions:
- Why is this non-linear?
- Would a single-layer perceptron work?
- How would the decision boundary differ from the checkerboard?

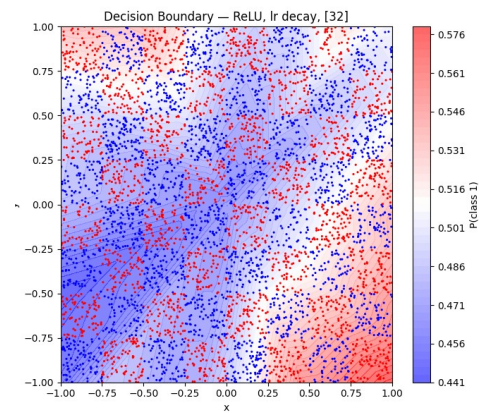
Experiment

- I asked Claude code to allow me to run multiple experiments with different activation functions for hidden layers and number of neurons per layer.
- Initially I used the 500 samples, but asked Claude to add the ability to have more samples.

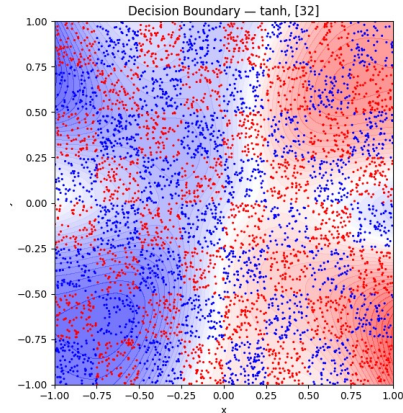
Running the tests with 64 Neurons.



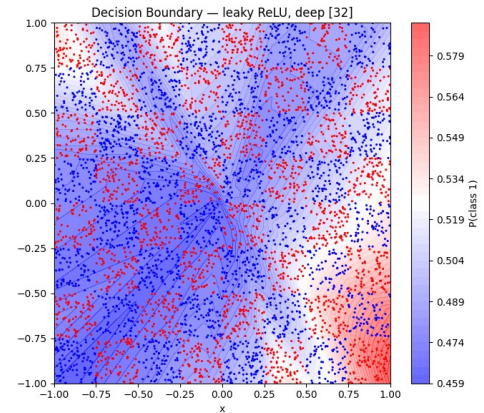
sigmoid



ReLU



tanh



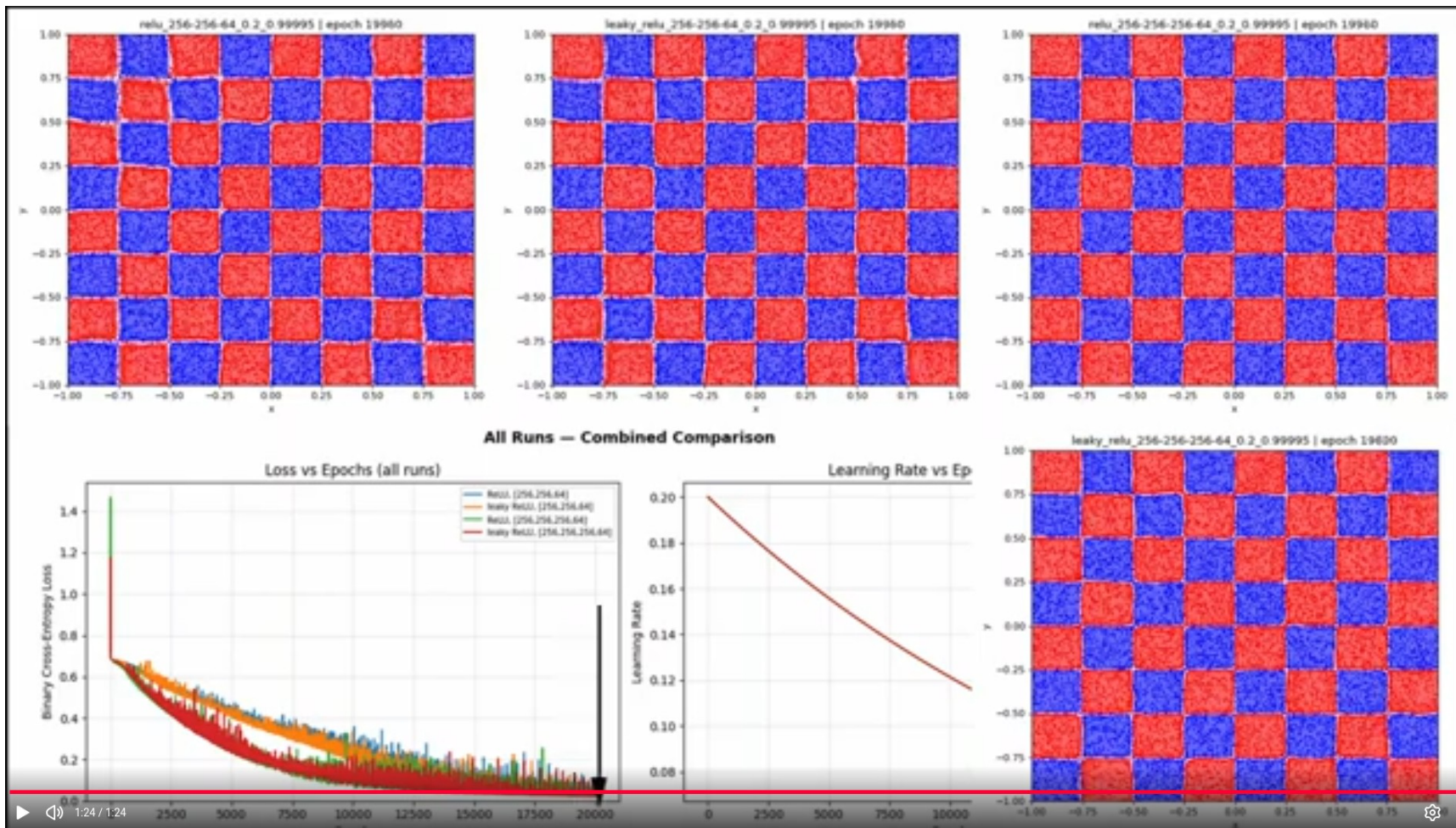
Leaky
ReLU

- Initial test were horrific - and I did not understand what was wrong.

Running a few more tests

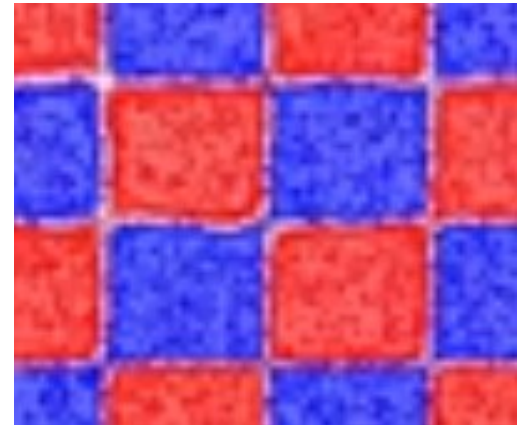
- I eventually figured out I needed many more neurons, data points, and epochs.
 - Try [256,256,64] 10000 points, 40000 epochs
 - Try [256,256,256,64] 10000 points, 40000 epochs
 - relu
- I was still thinking I needed different activation functions
-

Results — with [256,256,64],[256,256,256,64]



Created YouTube video

- <https://youtu.be/Xh22K4niArM>
- [256,256,256,64] learns the pattern in fewer epochs.
- [256,256,64] still has some areas that are incorrect.
- I believe it is the sample size



Extra Credit

1. Why is this non-linear?

- The boundary between the two (or more) patterns can not be drawn with a straight line

2. Would a single-layer perceptron work?

- No, a single layer can only make linear boundaries. But it can be made with fewer neurons and hidden layers

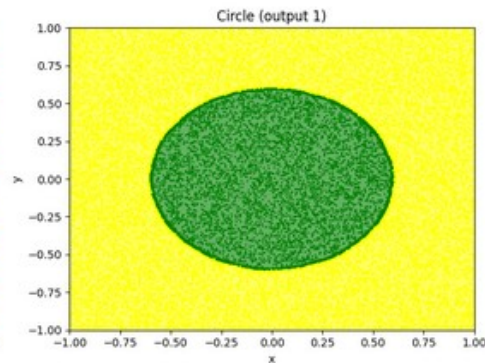
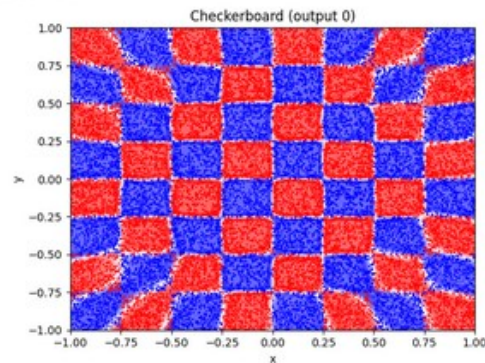
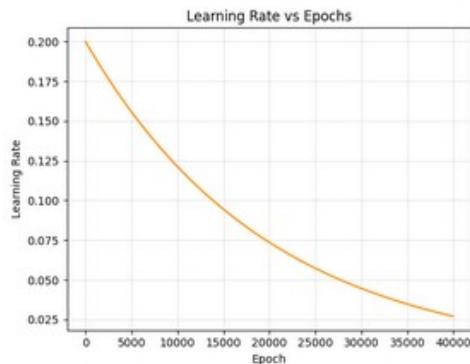
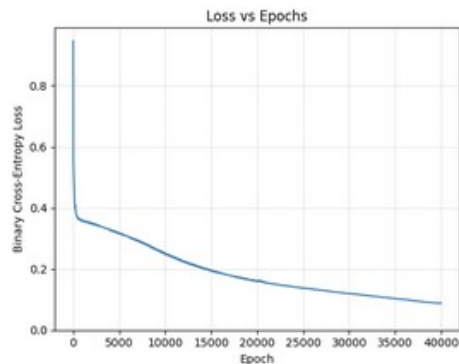
3. How would the decision boundary differ from the checkerboard?

- The checker board is complex with many regions. The circle only has two areas and one boundary to define as either in or out. It is much simpler.

Circle Results-[256,64] [256,256,64]

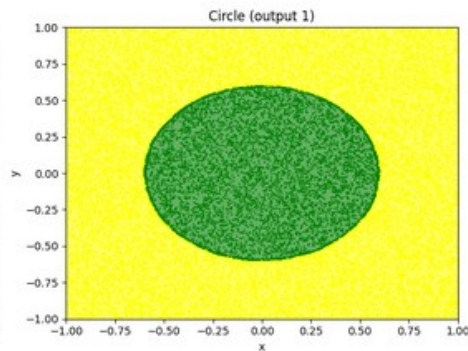
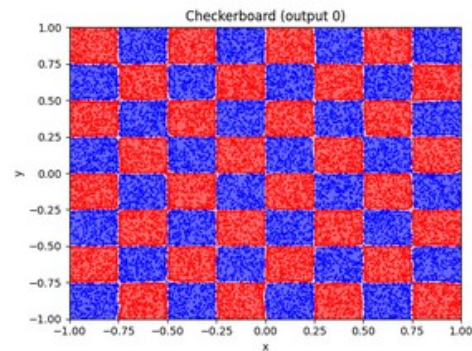
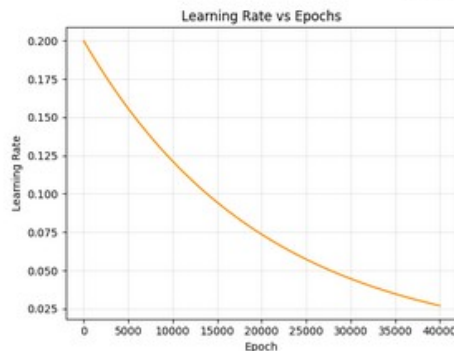
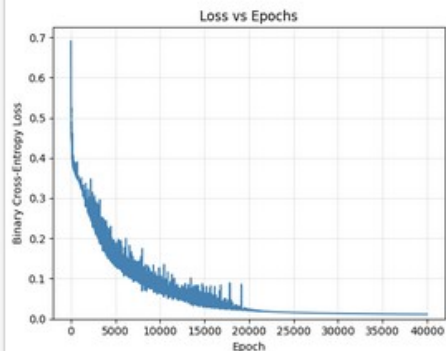
Final loss: 0.0879 | PNGs saved: 1600 checker + 1600 circle

ReLU, [256,64]

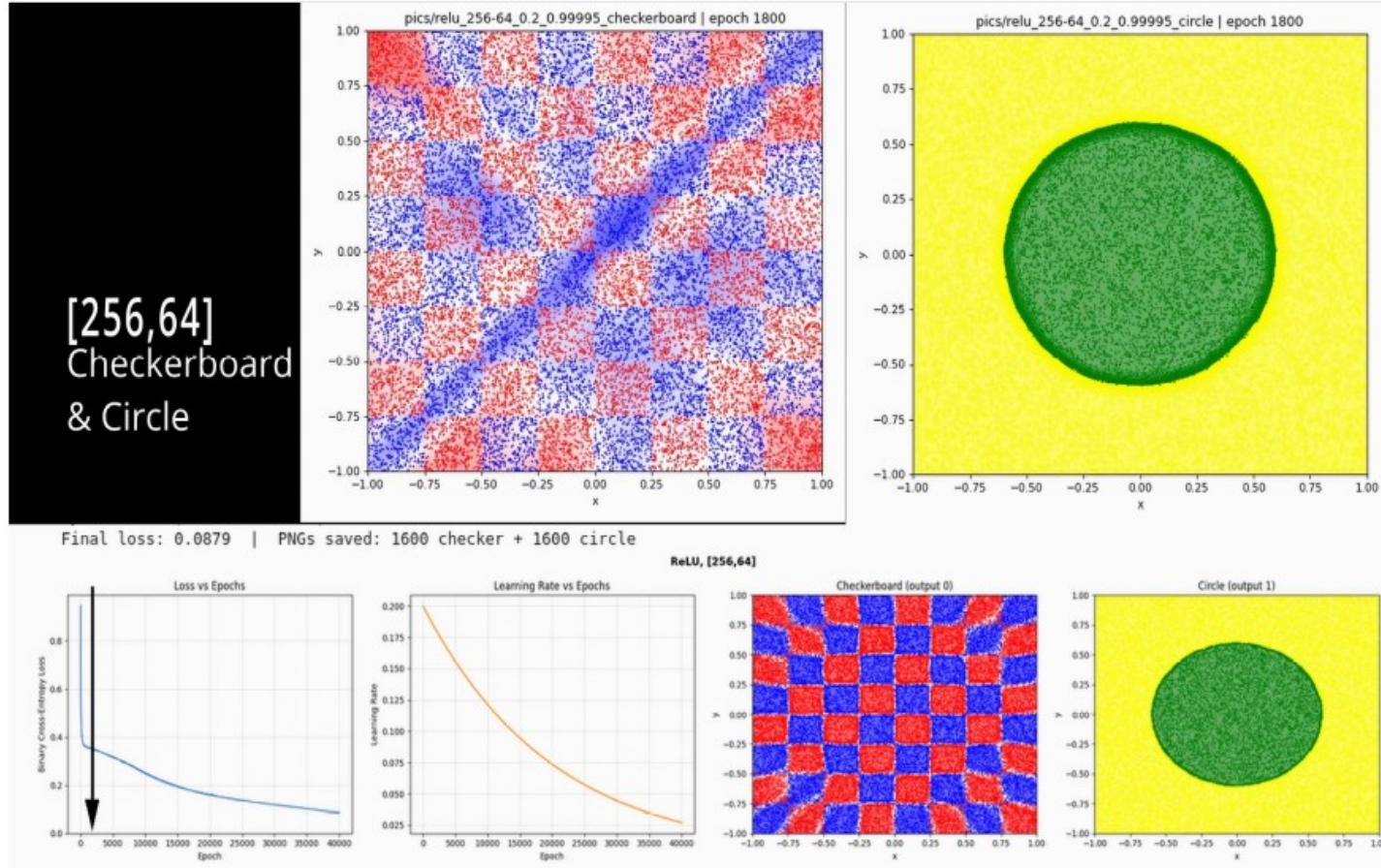


Epoch: 30000 | Loss: 0.0879 | PNGs saved: 1600 checker + 1600 circle
Final loss: 0.0114 | PNGs saved: 1600 checker + 1600 circle

ReLU, [256,256,64]



Youtube video - <https://youtu.be/Zjw3nQ-dbyU>



What did I learn

- The number of neurons is very important.
- The number of layers impacts the ability and speed of learning.
- Areas and nonlinearity affects the architecture needed.
- The number of epochs is important and relates to data set size and architecture
- Over training is an issue. The corners on my checkerboard are curved and it looks like this comes from my data set being sparse in that region.